PAPER_896: Phonon Modulation Factor 1.25THz Gaussian

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-08 **Session:** 209 **Source:** Sessions 204-208 standalone module integration **Calculator:** PhononModulationFactor125THzGaussianCalc (CP4 #480) **CVW:** v2.0.0 compliant

---

Abstract

Phonon modulation factor $\Phi\_\{1.25THz\}(\omega) = \Phi_0 \cdot \exp[-(\omega-\omega\_SCm)^2/(2\Gamma^2)] \cdot S_{26}$([SSq]) with quality factor Q = ω_SCm/Γ = 6.25 (sharp resonance) and FWHM = $2\Gamma\sqrt{}(2\ln 2) \approx 1.49$ THz linewidth. Characterizes the SCm phonon resonance profile for LENR coupling.

---

1. Core Equations

$$\Phi(\omega) = \Phi_0 \cdot \exp[-(\omega-\omega\_SCm)^2/(2\Gamma^2)] \cdot S_{26}$$
$$Q = \omega\_SCm/\Gamma$$
$$FWHM = 2\Gamma\sqrt{}(2ln2)$$

---

2. Parameters

Parameter	Default	Description
omega	$2\pi\times1.25e12$ rad/s	Angular frequency

---

3. UQFF Integration

This calculator integrates et_phonon_resonance.py from Session 208 into the CP4 pipeline as class #480.

PAPER_896 — Star Magic UQFF Framework — CVW v2.0.0